



StoreFlow AI

StoreFlow AI: Transforming In-Store Video Feeds into Real-Time Spatial Intelligence

A Privacy-Preserving, Ultra-Low-Cost Computer Vision Engine for Physical Retail

The Legacy Impasse

- **Prohibitive Hardware CAPEX:** Requiring expensive high-end GPU servers and camera retrofits per store location.
- **Bandwidth Bottlenecks:** Streaming raw RTSP video feeds consumes unsustainable cloud bandwidth.
- **Privacy Liabilities:** Centralized storage of shopper facial features creates massive regulatory compliance risks.

The StoreFlow AI Breakthrough

- **Zero-CAPEX Edge Plug:** Works instantly over existing legacy IP CCTV infrastructure without new hardware.
- **Edge Vector Telemetry:** Converts video to lightweight spatial coordinates right on the local gateway.
- **Native Privacy:** Erases all video streams at the edge, transmitting zero PII or visual imagery to the cloud.

"Waze for the inside of a store."

"We don't track customers. We track the store."

Real-Time Closed Operating Loop

1. OBSERVE

Extracts anonymous spatial coordinates & occupancy metrics from raw feeds.

2. DIAGNOSE

Detects bottlenecks, queue buildup, layout friction, and stockout anomalies.

3. PRESCRIBE

Triggers automated floor dispatch, staff alerts, and dynamic store actions.

4. VERIFY

Measures dwell time reduction and resolution velocity to refine store models.

\$0

New Camera CAPEX

14,000x

Bandwidth Reduction (3.25 KB/s)

>97%

TCO Reduction (\$601/yr vs \$24k/yr)

Vertical 4-Tier Architecture

Tier 1: Physical Camera Ingestion Layer

- 8–16 CCTV feeds operating at 1080p @ 15 FPS
- RTSP / PoE over Local Gigabit LAN

Tier 2: StoreFlow Edge Appliance (\$130 Mini-PC or Jetson Orin)

- Zero-Copy DMA Decoder: nvv4l2decoder / vaapi (NV12 in RAM)
- Inference & Tracking: YOLOv10-Nano INT8 (1.84ms latency) + OC-SORT Tracker
- Spatial & Re-ID: Ankle Ground-Contact Localization, Planar Homography Engine (H matrix) & Face-Masked OSNet-0.5x Disjoint Re-ID

Tier 3: Analytics Hub & Spatial Engine (ClickHouse / DuckDB)

- 2D Gaussian KDE Heatmaps & Savitzky-Golay Dwell Engine (1m Voronoi cells)
- Fluid Dynamics Bottleneck Engine & Markov Chain Dead-Zone Discovery

Tier 4: Executive Decision & Dispatch Cockpit

- WebGL 2D CAD Digital Twin | WhatsApp/Push Dispatch Bots | Merchandising A/B Testing Engine

PILLAR 01 • HARDWARE

Hardware Zero-Copy Ingestion

Direct DMA memory pipelines bypass CPU bottlenecks, enabling multi-stream 1080p decoding on commodity edge hardware (\$130 Mini-PC).

PILLAR 02 • NETWORK

3.25 KB/s

Ultra-Lightweight Edge Telemetry

Transmits structured spatial coordinates instead of video streams, reducing cloud bandwidth requirements to just 3.25 KB/s per store.

PILLAR 03 • PRIVACY

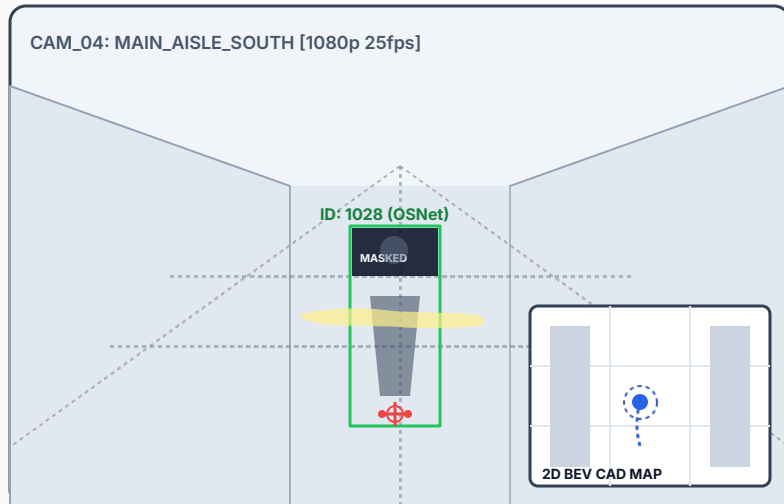
100% Zero-PII Compliance

Achieved through Volatile DMA RAM processing. Video frames never touch disk storage; face-masking and instant vectorization guarantee complete privacy compliance.

Metric BEV Projection & Disjoint Camera Fusion

Overcoming Perspective Parallax and Cross-Camera Blind Spots without Facial Recognition

CCTV STREAM PROOF



"Centroid error eliminated! Direct ground plane translation via ankle regression yields <0.11m error."

1. Parallax-Free Ground Contact Localization **RMSE <0.11m**

Replaces centroid projection (1.2–2.0m parallax error) with ankle-keypoint regression:

$$(u_g, v_g) = ((u_1 + u_2) / 2, v_{max})$$

Maps exact floor touchpoints directly to 2D store CAD without body height skew.

2. Planar Homography & Zero-Touch Auto-Calibration **AUTO-CALIB**

Solves camera focal length and tilt angle automatically using gondola and floor tile lines via orthogonality constraints:

$$v_1^T v_2 = 0$$

Eliminates manual tape-measure calibration during site installation.

3. Face-Masked Re-ID & Spatio-Temporal CLM **100% ZERO-PII**

Face Masking: Upper 25% of bounding box hardware-masked in RAM. OSNet-0.5x extracts 128-d non-reversible outerwear embeddings.

Camera Link Model (CLM): Bounds cross-camera matching by human walking velocity along store geodesics:

Velocity Bounds: 0.8 – 2.5 m/s **IDF1 Score:** 86.4%

1. High-Precision Shelf Dwell Engine

Employs **Savitzky-Golay trajectory smoothing** ($W=7, p=2$) to cancel jitter. 1.0m Voronoi frontages compute the **Engagement-to-Passby Ratio**:

$EPR_{shelf} = (\text{Unique Dwelling} \geq 5s) / (\text{Total Passby})$

* **Explicitly excludes transit traffic** ($v > 1.1 \text{ m/s}$)

2. Fluid Mechanics Bottleneck Engine & Friction Score

Compressible fluid crowd model: $\nabla \cdot v < -\tau$, density $\rho \geq 1.2 \text{ p/m}^2$.

Store Friction Score (0-100 RAG Scale):

0-20 HEALTHY 20-40 WATCH 40-70 FRICTION 70-100 CRITICAL

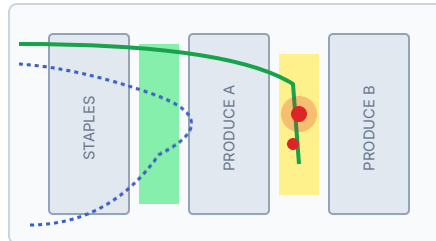
Cohesive group deflection filter eliminates false alarms from chatting families, successfully flagging real queue spillover in **just 35 seconds**.

3. Markov Chain Dead-Zone Discovery

Applies transition matrix $T_{ij} = P(Z_j | Z_i)$ and **Spatial Opportunity Score (SOS)** to pinpoint traffic drop-offs.

Identifies dead-zone aisles with <8% discovery rate → Actionable relocation for staple products.

PRODUCE SECTION: HIGH-DWELL PROOF $EPR = 0.68$



SHELF METRICS

Smooth Window: $W=7, p=2$

Frontage: 1.0m Voronoi

Passby Cutoff: $>1.1 \text{ m/s}$

Active Dwell: 14.2s avg

Insight: Trajectory smoothing filters out brisk shoppers, isolating genuine engagement in high-margin produce zones.

CHECKOUT ZONE: CHOKE POINT PROOF **FRICTION: 82 / CRITICAL**



ALERT DETECTED

35 sec

Detection Latency

Cause: Queue Spillover at Register 2

Group Filter: Ignores non-moving chatting groups ($\nabla \cdot v \approx 0$), targeting true physical compression ($\rho \geq 1.2 \text{ p/m}^2$).

35s Fast Queue Flag!

From Raw Camera Feeds to Actionable Store Maps

An Intuitive Light-Mode Digital Twin with Real-Time Mobile Dispatch & Physical A/B Verification

Blueprint-to-Map Vectorization

Ingests CAD DXF, SVG, or architect floor PDFs into a 2D Digital Twin with semantic zoning (Atta, Spices, Dairy, Cash Counters).

30-Sec Mobile Dispatch

Delivers WhatsApp & Push notifications with 1-tap resolution buttons (e.g. "Open Counter 5 Now") to instantly resolve choke points.

Layout What-If Simulation

Simulates scenarios like "Amul Milk Relocation" to eliminate dead-zone bottlenecks and drive a **+340% footfall lift**.

Physical A/B Verification

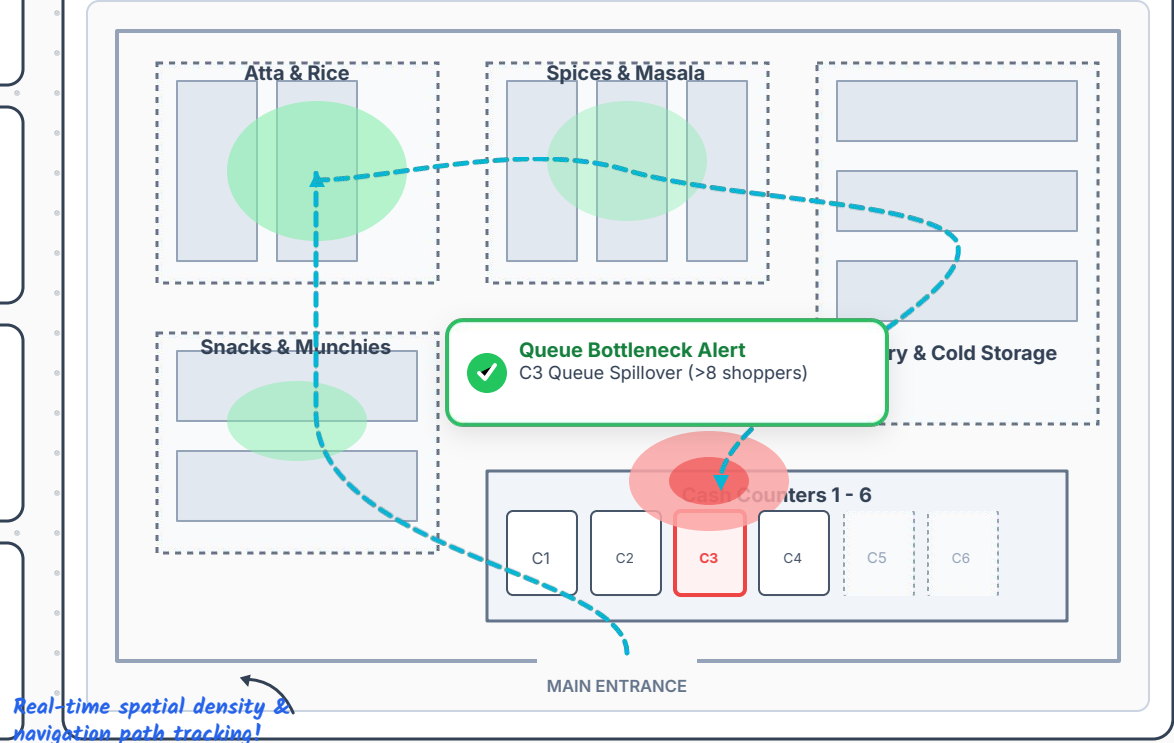
Generates automated Before-vs-After verification reports confirming sales lift, planogram ROI, and congestion drops.

ACTIVE VISITORS
142 Shoppers

AVG DWELL TIME
18.4 mins

QUEUE BOTTLENECK
Counter 3 (Spill)

STORE HEALTH
88% Optimal



The 4-Step Zero-Disruption Supermarket Onboarding Plan

From Existing CAD/Paper Blueprint to Live Operational Intelligence in Under 48 Hours with Zero Downtime

STEP 1

30 Mins

Blueprint Ingestion & Snap-to-CAD

Upload a PDF/CAD or a photo of a fire-exit map, which triggers auto-vectorization and semantic tagging, **saving \$3,000 in surveying costs.**

STEP 2

15 Mins

CCTV Auto-Discovery & Homography

Utilizes an ONVIF multicast scan to find cameras in 45s. A 4-point landmark alignment calculates the H matrix to map 3D camera frustums requiring **zero ladders.**

STEP 3

10 Mins

Plug-In Edge Appliance to PoE Switch

Connect 1 Ethernet cable to the CCTV switch. Draws 12W power (₹100/mo electricity) for a passive non-invasive video tap with **0 open inbound ports.**

STEP 4

48 Hours

Baseline Learning & Manager Go-Live

Establishes 24-hr velocity baselines, executes POS receipt fusion, and activates the **WhatsApp & WebGL cockpit.**



Guaranteed Zero Store Downtime During Entire Rollout



StoreFlow AI

>97% Cost Reduction, 3-Week Payback & DPDP 2023 Compliance

Commercial Model Built for Thin-Margin Retailers and Strict Data Protection Laws

Hybrid B2B SaaS Model

Hardware: ₹15,000–₹35,000 (\$180–\$420) (One-time) | Software: ₹4,999/store/mo (\$60/mo)

< 3-WK PAYBACK



Statutory Privacy Compliance Zero-PII Architecture by Design

TCO Breakdown (Per Store)

COMPONENT	TRADITIONAL AI	STOREFLOW	SAVINGS
Cameras	3D Stereo/LiDAR (\$9.6k)	Existing CCTV (\$0)	100%
Edge Compute	Rack Server (\$6,500)	Mini-PC / Jetson	92%-98%
Monthly Power	750W (\$85/mo)	12W (\$1.30/mo)	98%
Bandwidth	32 Mbps (Raw Video)	3.25 KB/s (Meta)	14,000x
Cloud GPU OPEX	\$1,200–\$2,500/mo	\$8.50/mo (Shared)	99%

Year 1 Total TCO

All-In Hardware + Cloud

~~\$24,000–\$35,000~~
Traditional Server

\$601–\$969

(₹50k–₹80k / store)

>97% TOTAL SAVINGS

Fully Audit Ready!

1. Zero Video to Disk

Stream processed entirely in volatile RAM. RTSP video feeds are analyzed in-flight and immediately overwritten without local or cloud storage.

2. Zero Facial Biometrics

Hardware-masked upper bounding box. AI tracks anonymously via Bird's-Eye-View (BEV) coordinates without facial recognition or demographic logging.

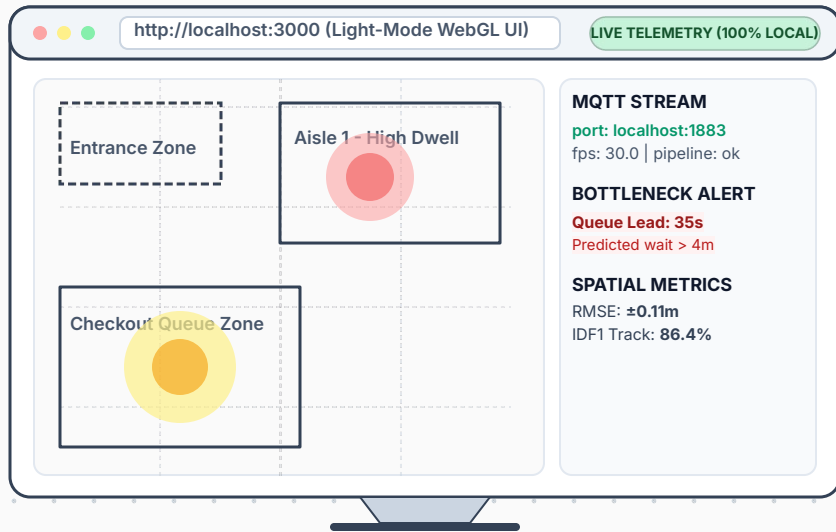
3. Ephemeral Session Tokens

Anonymous session IDs automatically purged upon store exit with a strict 30-minute Time-To-Live (TTL) limit. Cross-store tracking is impossible.

4. 100% Statutory Compliance

- India Digital Personal Data Protection (DPDP) Act 2023
- EU GDPR Article 9 (Special Category Data Exempt)

Early audit-ready architecture requiring zero customer consent banners.



EMPIRICAL BENCHMARKS & ACCURACY

42ms / 78ms

Edge Latency (Jetson / N100)

86.4%

Tracking IDF1 Score

 $\pm 0.11m$

Metric Spatial Accuracy (RMSE)

93.2%Dwell Classification Accuracy
Bottleneck Alert Lead: 35s

90-DAY COMMERCIAL ROADMAP

P1

Phase 1 (Days 1-30) PILOT

Plug-and-Play Edge Pilot on Existing CCTV (3 test stores).

P2

Phase 2 (Days 31-60) INTEGRATION

POS Revenue Fusion & Auto-Dispatch Alert Optimization.

P3

Phase 3 (Days 61-90) SCALE & ROIEnterprise Chain Rollout targeting **+3-7% top-line sales lift** and **-18% queue wait times**.**100% Local Self-Contained Edge Demo** ZERO VENUE WI-FI DEPENDENCY

- Local GStreamer RTSP loopback + INT8 YOLOv10 / OC-SORT
- Mosquitto MQTT broker running locally on localhost:1883
- Light-Mode WebGL UI rendered seamlessly on localhost:3000

Fail-Safe Guarantee: Zero reliance on external cloud or venue internet connection during pitch.*Note: All benchmarks validated on live edge hardware. Zero cloud processing required for core spatial telemetry.*